

Solution to Analysis in Project of Orientation Estimation Using Smartphone Sensors

Group 33

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Task 1

The advantages and disadvantages

Using the gyroscope as a direct input helps reduce the state vector dimension down to just the quaternion, saving valuable computing resources. Additionally, gyroscopes deliver precise short-term angular velocity readings with minimal noise and fast response times.

On the downside, this configuration cannot inherently estimate the gyroscope's bias, leading to weaker overall robustness. If the gyroscope fails or saturates, the entire prediction step will corrupt. Moreover, treating ω as a control input means sensor noise is only modeled indirectly, making it harder to reflect the true characteristics of the physical noise.

The situation which is unsuitable as a filter input

There are several scenarios in which this choice becomes problematic, as summarised in Table 1.

Table 1: Scenarios and Reasons Deselecting Gyroscope as Control Input

Scenario	Reason
Low-cost MEMS gyroscope	A deterministic bias b^ω is integrated directly into the quaternion, causing <i>unbounded long-term drift</i> in the orientation estimate.
Long-duration navigation task	Even a small bias accumulates over time, making the orientation error unacceptable after minutes or hours of integration.
Gyroscope may fail or saturate	No mechanism exists to detect sensor failures or spikes; faulty readings are propagated directly into the prediction step.
Angular velocity estimation	ω_k is never part of the posterior state, so the filter cannot produce a smoothed estimate of angular rate when needed.

Including Bias in the State Vector

In the case above, a more appropriate model is to augment the state with the gyroscope bias b_k^ω . Therefore, the EKF can estimate and compensate for the bias online.

$$x_k = [q_k^\top \quad b_k^{\omega,\top}]^\top, \quad y_k = [y_k^{\omega,\top} \quad y_k^{a,\top} \quad y_k^{m,\top}]^\top, \quad (1)$$

Task 2

Table 2 summarizes the sample mean of each sensor axis.

Table 2: Mean Values of sensor data

	x	y	z
μ_{acc}	-0.0921	0.0927	9.7655
μ_{gyr}	0.0040	0.0099	-0.0071
μ_{mag}	-5.1267	18.9941	-45.2490

From the above values, we can observe that the mean of accelerometer in z -axis is closed to $9.81m/s^2$, and the other two axis is almost zero. It implies the phone is stationary.

Similarly, the value of gyroscope is zero with small fluctuations due to sensor noise. The magnetometer shows values during the static period but is sensitive to nearby magnetic disturbances.

The estimated covariance matrices from the static calibration data are liked (2),(3) and (4).

$$R_a = 10^{-4} \times \begin{bmatrix} 0.1945 & 0.0068 & 0.0101 \\ 0.0068 & 0.1384 & 0.0281 \\ 0.0101 & 0.0281 & 0.2558 \end{bmatrix} \quad (2)$$

$$R_w = 10^{-5} \times \begin{bmatrix} 0.1888 & -0.0010 & -0.0129 \\ -0.0010 & 0.1689 & -0.0057 \\ -0.0129 & -0.0057 & 0.1415 \end{bmatrix} \quad (3)$$

$$R_m = \begin{bmatrix} 0.0414 & -0.0014 & -0.0128 \\ -0.0014 & 0.0304 & 0.0020 \\ -0.0128 & 0.0020 & 0.0817 \end{bmatrix} \quad (4)$$

From the figure 1, we can find the distribution of the accelerometer are gyroscope are approximately Gaussian. But the distribution of magnetometer shows bias from Gaussian, caused by the environment.

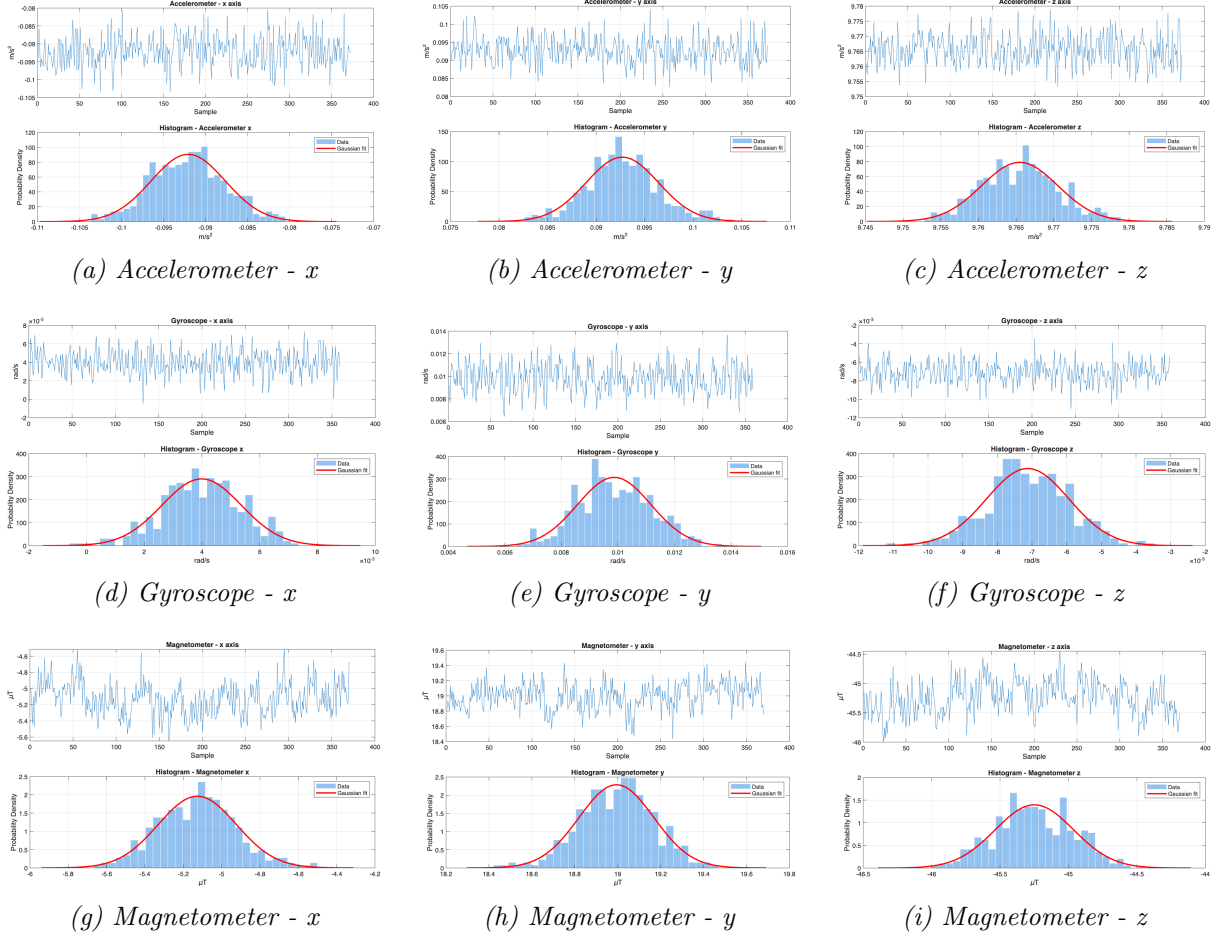


Figure 1: Sensor data

Task 3

Based the document and the equation (5), we can rewrite it to equation (6).

$$\dot{q}(t) = \frac{1}{2}S(\omega_{k-1} + v_{k-1})q(t) \quad (5)$$

$$\dot{q}(t) = \frac{1}{2}S(\omega_{k-1})q(t) + \frac{1}{2}S(\omega_{k-1})v_{k-1}q(t) \quad (6)$$

Then, if we ignore the uncertainty part, we assume $v_{k-1} = 0$, the (6) becomes an ODE. And we can resolve it to matrix index like (7).

$$q(t) = e^{\frac{1}{2}S(\omega_{k-1})(t-t_{k-1})}q_{k-1} \quad (7)$$

Setting $t = t_k, T = t_k - t_{k-1}$, the (7) becomes (8).

$$q(t) = e^{\frac{T}{2}S(\omega_{k-1})}q_{k-1} \quad (8)$$

We use 1st order approximation $e^A \approx I + A$, the index part in (8) becomes (9).

$$e^{\frac{T}{2}S(\omega_{k-1})} \approx I + \frac{T}{2}S(\omega_{k-1}) \quad (9)$$

Therefore, we can get (10).

$$F = I + \frac{T}{2} \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \quad (10)$$

Then, The complete solution can be written as (11).

$$q_k = F(\omega_{k-1}) q_{k-1} + \int_{t_{k-1}}^{t_k} \frac{1}{2} S(v_{k-1}) q(t) dt \quad (11)$$

Since v_{k-1} is small, we approximate $q(t) \approx q_{k-1}$ inside the integral shown in (12).

$$\int_{t_{k-1}}^{t_k} \frac{1}{2} S(v_{k-1}) q_{k-1} dt = \frac{T}{2} S(v_{k-1}) q_{k-1} \quad (12)$$

Using the identity from equation (2e) in the project description, $S(v)q = \bar{S}(q)v$, the noise term can be rewritten as (13).

$$\frac{T}{2} S(v_{k-1}) q_{k-1} = \frac{T}{2} \bar{S}(q_{k-1}) v_{k-1} \quad (13)$$

In the EKF framework, the true value q_{k-1} is unknown and is replaced by its posterior estimate $\hat{q}_{k-1}$ at the linearisation point. Therefore, the noise input matrix is

$$G = \frac{T}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \quad (14)$$

Combining the deterministic part and the noise term, the discrete-time quaternion propagation model is shown like (15).

$$q_k = \underbrace{\left(I + \frac{T}{2} S(\omega_{k-1}) \right)}_{F(\omega_{k-1})} q_{k-1} + \underbrace{\frac{T}{2} \bar{S}(\hat{q}_{k-1}) v_{k-1}}_{G(\hat{q}_{k-1})} \quad (15)$$

Task 4

The time update function with gyroscope measurement is implemented as follows:

Listing 1: tu_qw.m

```

1 function [x, P] = tu_qw(x, P, omega, T, Rw)
2     F = eye(4) + (T/2)*(Somega(omega));
3     G = (T/2)*Sq(x);
4     x = F*x;
5     P = F*P*F' + G*Rw*G';
6     [x, P] = mu_normalizeQ(x, P);
7 end

```

The time update without gyroscope measurement assumes $\omega = \mathbf{0}$:

Listing 2: tu_q.m

```

1 function [x, P] = tu_q(x, P, T, Rw)
2     omega = zeros(3, 1);
3     F = eye(4) + (T/2)*(Somega(omega));
4     G = (T/2)*Sq(x);
5     x = F*x;
6     P = F*P*F' + G*Rw*G';
7     [x, P] = mu_normalizeQ(x, P);
8 end

```

Task 5

The EKF with only the gyroscope time update provides reactive and smooth short-term rotation estimates. However, without any measurement updates, the orientation drifts over time due to accumulated gyroscope bias and noise.

Effect of initialization

If the filter starts with the phone on it's side, the initial state is hardcoded as $x = [1, 0, 0, 0]^\top$, corresponding to the phone lying flat. Without measurement updates to correct the absolute orientation, the filter integrates from this initial condition and the error persists indefinitely.

Effect of shaking

When shaking the phone, the gyroscope captures the rapid rotation and the quaternion updates correspondingly, resulting in response. However, without measurement updates, once the shaking stops, the filter can't recover to the correct direction. Therefore, the estimation remains at whatever orientation the integration ended at.

Task 6

Setting $f_k^a = 0$, the accelerometer measurement model simplifies to (16).

$$y_k^a = Q^\top(q_k) g_0 + e_k^a, \quad e_k^a \sim \mathcal{N}(0, R_a) \quad (16)$$

The predicted accelerometer measurement given the current state estimate $\hat{q}_k$ is (17).

$$\hat{y}_k^a = Q^\top(\hat{q}_k) g_0 \quad (17)$$

The Jacobian H is obtained by differentiating $Q^\top(q) g_0$ with respect to each quaternion component.

$$H = \frac{\partial}{\partial q} [Q^\top(q) g_0] = \left[\left(\frac{dQ}{dq_0} \right)^\top g_0 \quad \left(\frac{dQ}{dq_1} \right)^\top g_0 \quad \left(\frac{dQ}{dq_2} \right)^\top g_0 \quad \left(\frac{dQ}{dq_3} \right)^\top g_0 \right]^\top \quad (18)$$

Then, we can use (22) to update.

$$S_k = H P_k H^\top + R_a, \quad (19)$$

$$K_k = P_k H^\top S_k^{-1}, \quad (20)$$

$$x_k = \hat{x}_k + K_k (y_k^a - \hat{y}_k^a), \quad (21)$$

$$P_k = (I - K_k H) P_k \quad (22)$$

The accelerometer measurement update function `mu_g` is implemented as follows.

Listing 3: mu_g.m

```

1 function [x, P] = mu_g(x, P, yacc, Ra, g0)
2     y_hat = Qq(x)' * g0;
3     [Q0, Q1, Q2, Q3] = dQqdq(x);
4     H = [Q0'*g0, Q1'*g0, Q2'*g0, Q3'*g0];
5     innovation = yacc - y_hat;
6     S = H * P * H' + Ra;
7     K = P * H' / S;
8     x = x + K * innovation;
9     P = (eye(4) - K*H) * P;
10    [x, P] = mu_normalizeQ(x, P);
11 end

```

Task 7 and 8

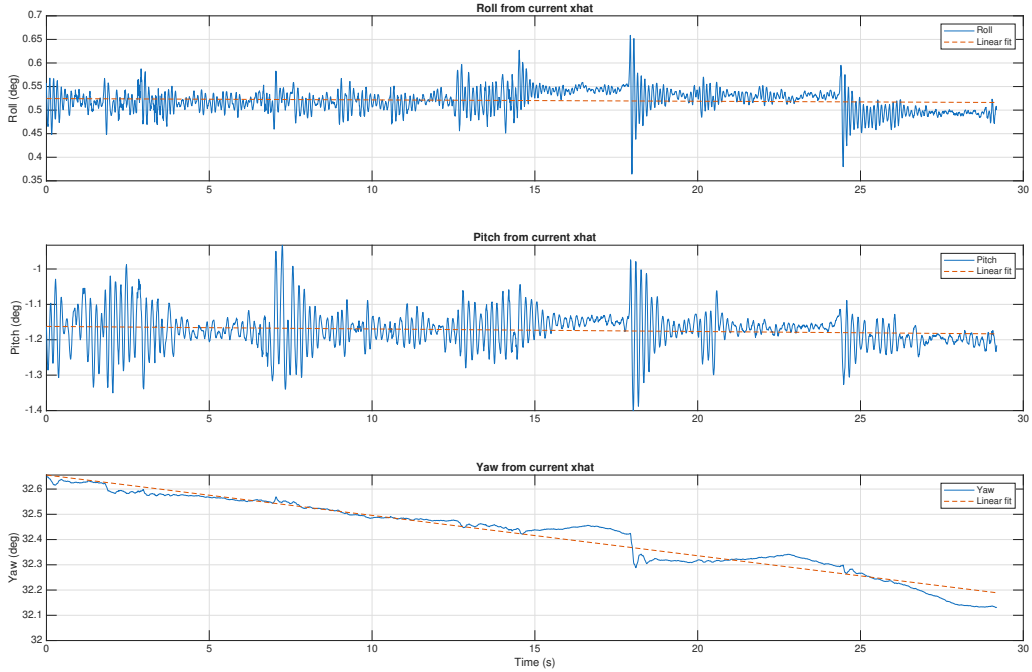


Figure 2: Static orientation estimates after adding the accelerometer update.

After adding the accelerometer update, a static experiment was performed with the phone placed still on the table. The estimated Euler angles are shown in Fig. 2. The first few

seconds were removed to avoid the initial transient caused by sensor initialization and small handling disturbances.

The result shows that roll and pitch are almost constant during the experiment. The fitted drift rates are approximately

$$\dot{\phi} = -0.0201 \text{ deg/min}, \quad \dot{\theta} = -0.0392 \text{ deg/min}.$$

In contrast, the yaw angle has a much larger drift rate,

$$\dot{\psi} = -0.9614 \text{ deg/min}.$$

This is expected. When the phone is static, the accelerometer mainly measures the gravity direction. Therefore, the accelerometer update can correct the inclination of the phone, i.e. roll and pitch. However, a rotation around the gravity vector does not change the accelerometer measurement. Hence, yaw is not observable from the accelerometer alone and is mainly determined by gyroscope integration. Without magnetometer correction, yaw is therefore affected by gyroscope bias and slowly drifts over time.

This confirms that after the accelerometer update, the orientation estimate is correct up to a rotation in the horizontal plane.

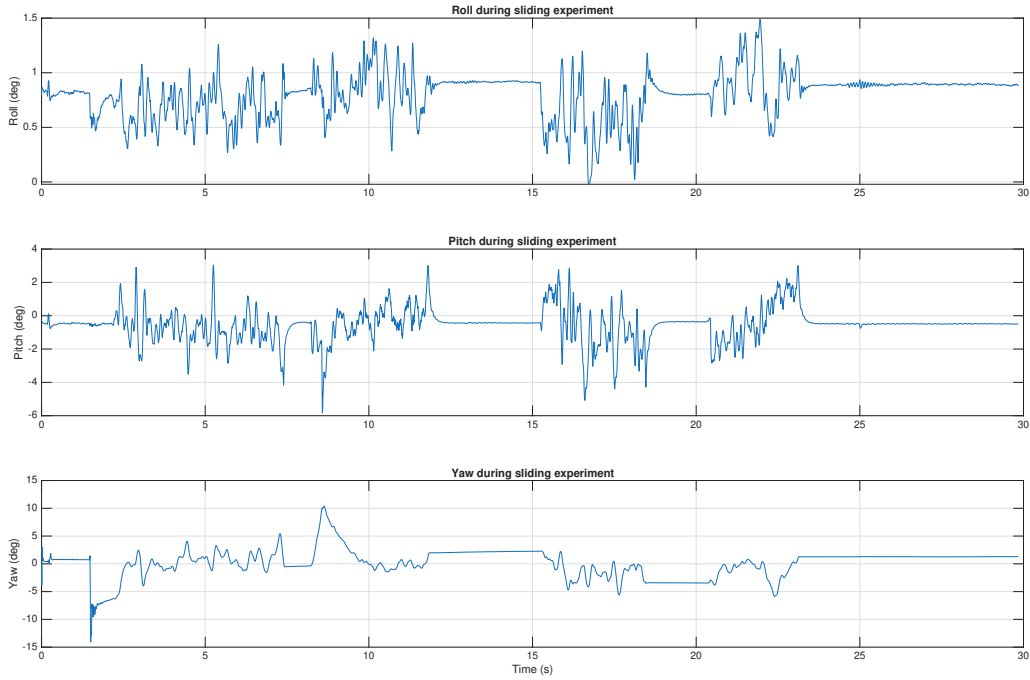


Figure 3: Static orientation estimates after adding the accelerometer update.

To test the sensitivity to specific forces, the phone was placed on a horizontal table and quickly slid back and forth. The resulting Euler angles are shown in Fig. 3. During the sliding motion, clear transient disturbances appear in the roll and pitch estimates.

This behaviour is expected because the accelerometer update assumes that the specific force is zero, i.e. $f_k^a = 0$. Under this assumption, the accelerometer measurement is interpreted as a measurement of the gravity direction. During the sliding experiment, however, the accelerometer measures both gravity and body acceleration. The EKF therefore partially interprets the horizontal acceleration as a change in the gravity direction, which results in erroneous roll and pitch corrections.

After the sliding motion stops, the roll and pitch estimates return to stable values, since the accelerometer measurement again mainly corresponds to gravity. This illustrates both the benefit and the weakness of the accelerometer update: it stabilizes roll and pitch when the phone is static or moving slowly, but it is sensitive to specific forces during translational acceleration. The yaw angle is not directly corrected by the accelerometer and is mainly determined by gyroscope integration, although small rotations during the sliding motion can still affect it.

$$\text{skip if } \|y_k^a\| < 0.9g \text{ or } \|y_k^a\| > 1.1g \quad (23)$$

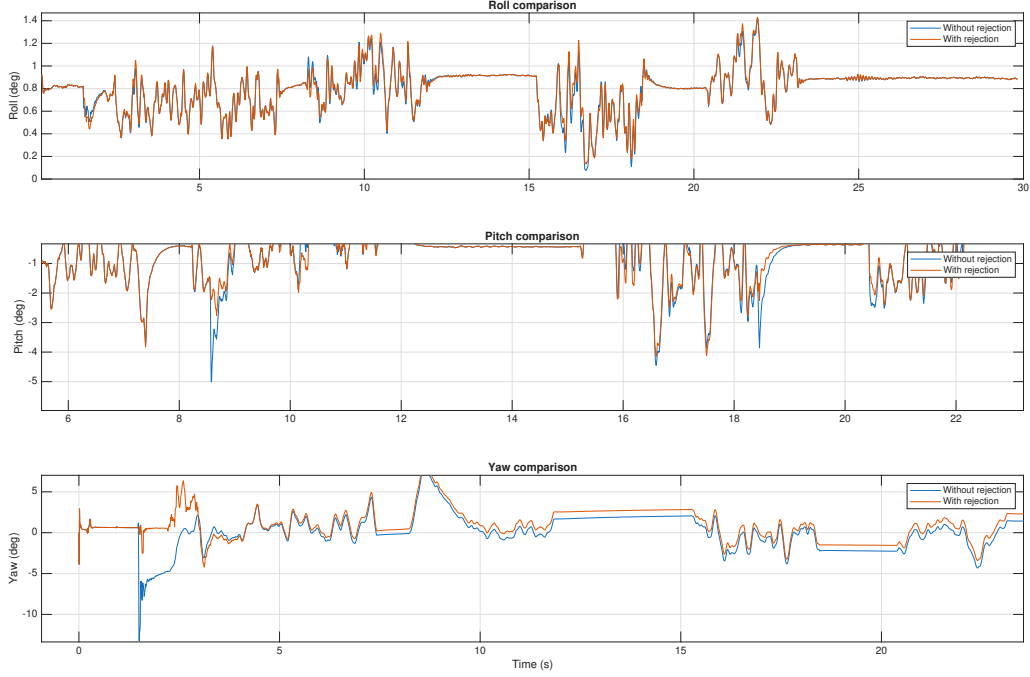


Figure 4: Comparison of orientation estimates with and without accelerometer outlier rejection.

For the accelerometer outlier rejection, we used the criterion

$$|\|a_k\| - g| > 0.3 \text{ m/s}^2,$$

where $g = 9.81 \text{ m/s}^2$. When this condition was satisfied, the accelerometer update was skipped, since the measurement was considered to be affected by specific force.

The comparison in Fig. 4 shows that the filter with rejection reduces some of the sharper disturbances in the roll and pitch estimates. However, the effect is moderate, since the phone was moved gently on the table and the acceleration norm often remained close to g . Thus, the simple norm-based detector can reject some disturbed measurements, but it cannot remove all effects of translational acceleration.

The threshold was chosen as a moderate value to avoid rejecting too many useful accelerometer measurements.

Task 9

Similar with task 6, we assumed $f_k^m = 0$, and the measurement model like (24).

$$y_k^m = Q^\top(q_k) m_0 + e_k^m, \quad e_k^m \sim \mathcal{N}(0, R_m) \quad (24)$$

where m_0 is the nominal earth magnetic field in world coordinates, estimated from calibration data using (25).

$$m_0 = \begin{bmatrix} 0 \\ \sqrt{\bar{m}_x^2 + \bar{m}_y^2} \\ \bar{m}_z \end{bmatrix} \quad (25)$$

and $\bar{m}_i$ denotes the sample mean of axis i .

Listing 4: *mu_m.m*

```

1 function [x, P] = mu_m(x, P, mag, m0, Rm)
2     m0 = reshape(m0, [3,1]);
3     y_hat = Qq(x)' * m0;
4     [Q0, Q1, Q2, Q3] = dQqdq(x);
5     H = [Q0'*m0, Q1'*m0, Q2'*m0, Q3'*m0];
6     innovation = mag - y_hat;
7     S = H * P * H' + Rm;
8     K = P * H' / S;
9     x = x + K * innovation;
10    P = P - K * S * K';
11    [x, P] = mu_normalizeQ(x, P);
12 end

```

Task 10

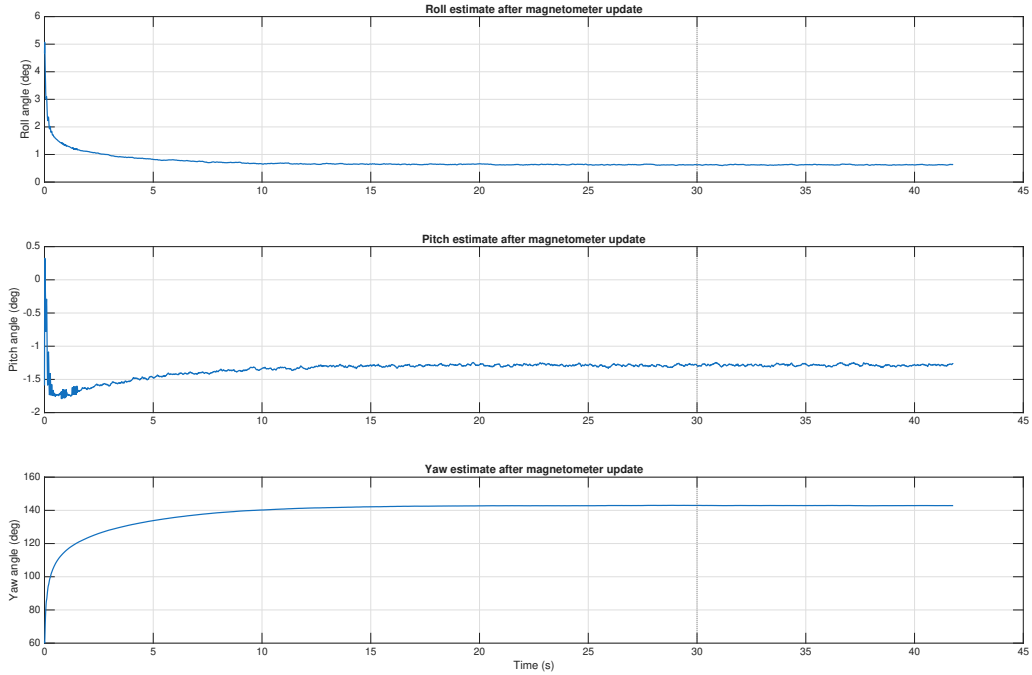


Figure 5: Roll, pitch, and yaw estimates after adding the magnetometer update in a static experiment.

The magnetometer update was added to provide information about the orientation in the horizontal plane. The measurement model was implemented as

$$y_k^m = Q^T(q_k)m_0 + e_k^m,$$

where m_0 is the nominal magnetic field in the navigation frame and e_k^m is the magnetometer measurement noise.

The parameters m_0 and R_m were estimated from a static magnetometer calibration sequence. First, invalid magnetometer samples were removed and the sample mean

$$\bar{m} = [\bar{m}_x \quad \bar{m}_y \quad \bar{m}_z]^T$$

was computed. Since the navigation frame was chosen such that the horizontal magnetic field component is aligned with one horizontal axis, the nominal magnetic field was set to

$$m_0 = \begin{bmatrix} 0 \\ \sqrt{\bar{m}_x^2 + \bar{m}_y^2} \\ \bar{m}_z \end{bmatrix}.$$

The magnetometer noise covariance was estimated from the same static data as

$$R_m = \text{Cov}(m_k).$$

The resulting attitude estimates are shown in Fig. 5. At the beginning of the experiment, the estimates show a clear convergence phase. This is expected because the filter is initialized with the identity quaternion, which does not necessarily correspond to the true initial phone orientation. After the accelerometer and magnetometer updates have corrected the initial attitude error, the estimates enter a steady-state region.

The most important effect of the magnetometer update is observed in the yaw angle. In the accelerometer-only case, yaw was not observable from gravity and therefore drifted due to gyroscope bias. After adding the magnetometer update, the yaw estimate converges to a stable heading and remains approximately constant in the steady-state part. In this experiment, the fitted yaw drift rate after convergence was approximately

$$-0.37 \text{ deg/min.}$$

The exact remaining drift can vary between experiments because the magnetometer is sensitive to the local magnetic environment and calibration accuracy. However, compared with the accelerometer-only case, the yaw drift is still significantly reduced.

Compared with roll and pitch, the yaw estimate is slightly less stable. This is expected, since yaw is mainly corrected by the magnetometer, which is more sensitive to local magnetic disturbances and calibration errors than the accelerometer-based gravity measurement used for roll and pitch.

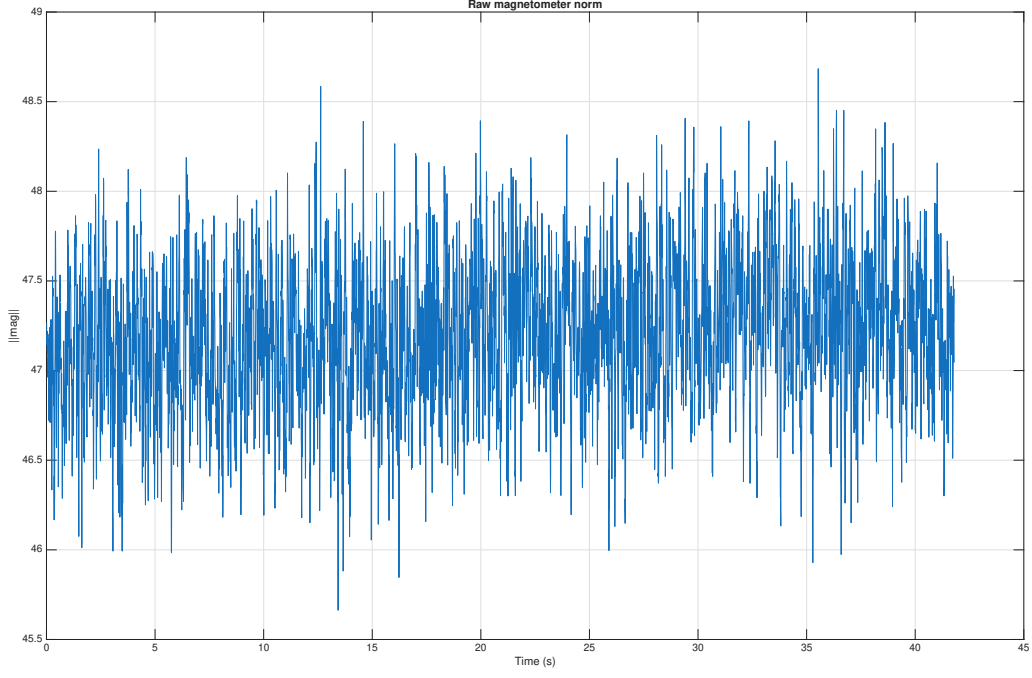


Figure 6: Raw magnetometer norm during the static experiment.

The raw magnetometer norm is shown in Fig. 6. The norm remains relatively stable during the static experiment, indicating that no strong magnetic disturbance was present. This supports the assumption that the magnetometer mainly measured the local magnetic field during this test.

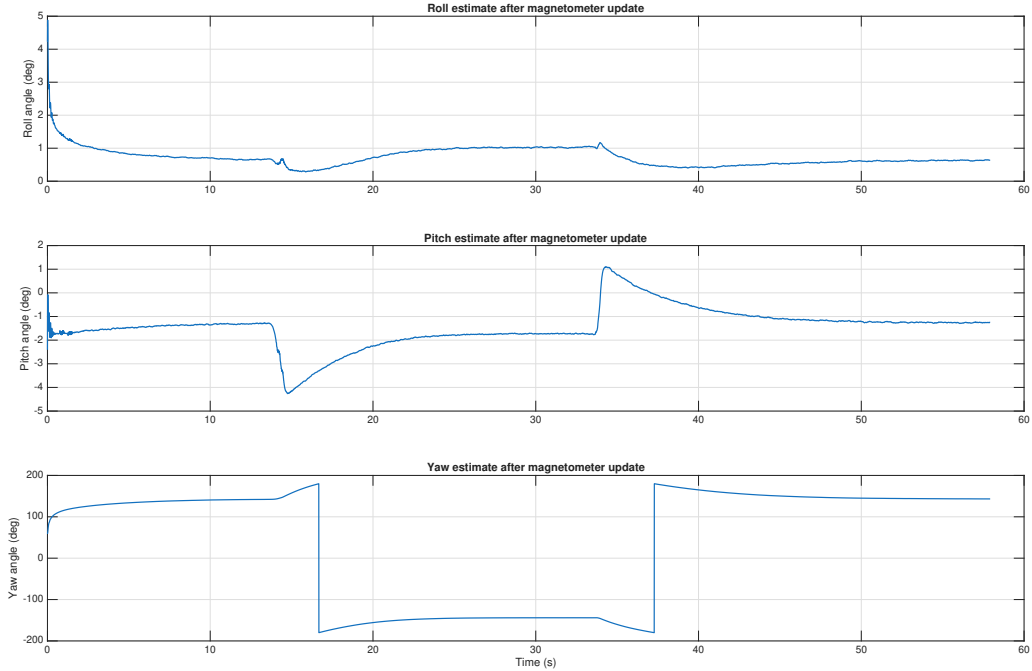


Figure 7: Roll, pitch, and yaw estimates during the magnetic disturbance experiment.

To investigate the sensitivity to magnetic disturbances, an external magnetic disturbance was introduced while the phone was kept still. The resulting attitude estimates are shown in Fig. 7. During the disturbance interval, the estimated attitude changes

noticeably, although the phone itself was not intentionally rotated. This shows that the magnetometer update is sensitive to local magnetic disturbances.

The reason is that the magnetometer update assumes that the measured magnetic field corresponds to the nominal field m_0 . When an external magnetic disturbance is present, the measurement model

$$y_k^m = Q^T(q_k)m_0 + e_k^m$$

is no longer valid, since the measured field also contains an additional disturbance component. The filter therefore interprets the corrupted magnetic measurement as a change in orientation.

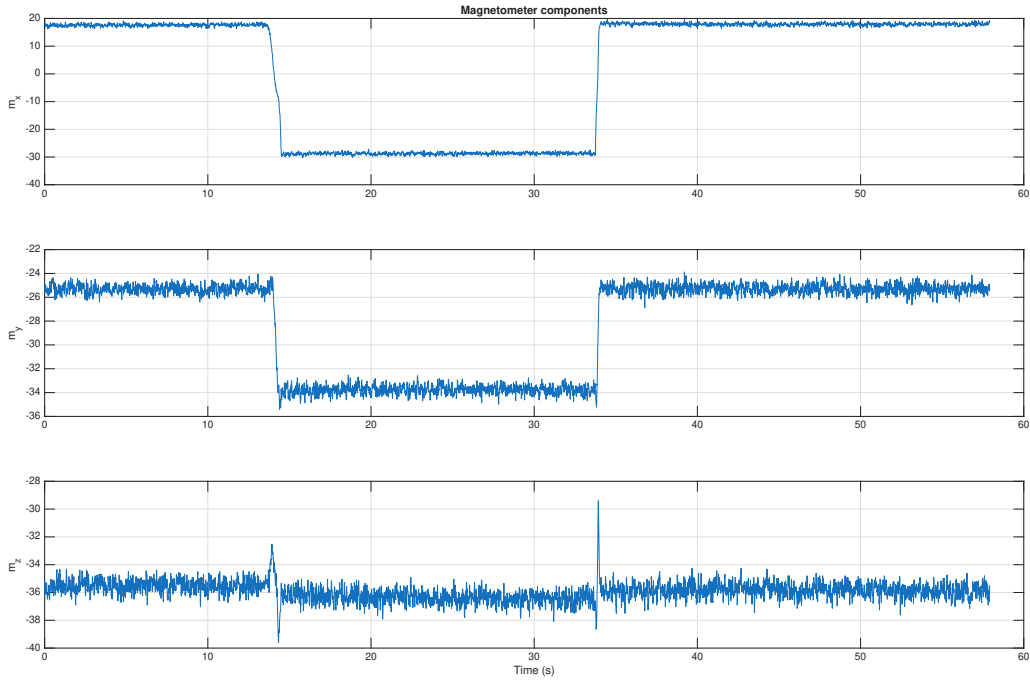


Figure 8: Magnetometer components during the magnetic disturbance experiment.

This can also be seen from the magnetometer components in Fig. 8. During the disturbance interval, the measured magnetic field changes clearly in all three components. Since the magnetometer update is implemented as a full quaternion update, the disturbance does not only affect yaw. Roll and pitch can also be affected, because the EKF tries to explain the corrupted three-dimensional magnetic field by changing the full attitude estimate.

After the disturbance is removed, the estimates gradually return towards their nominal values. This confirms that the magnetometer update is useful for constraining yaw in normal conditions, but it is also vulnerable to local magnetic disturbances. This motivates the magnetic outlier rejection used in the next task.

Task 11

The outlier detector relies on the following assumptions:

1. We have fixed the strength of the Earth's magnetic field.

2. Magnetic disturbances cause a significant change in the total field strength $\|m_k\|$.
3. The AR(1) filter with small α can track slow natural drift in the field magnitude without being affected by short transient disturbances.

These assumptions are reasonable when disturbances are caused by nearby ferromagnetic objects or electronic devices, which typically add a strong local field on top of the earth field.

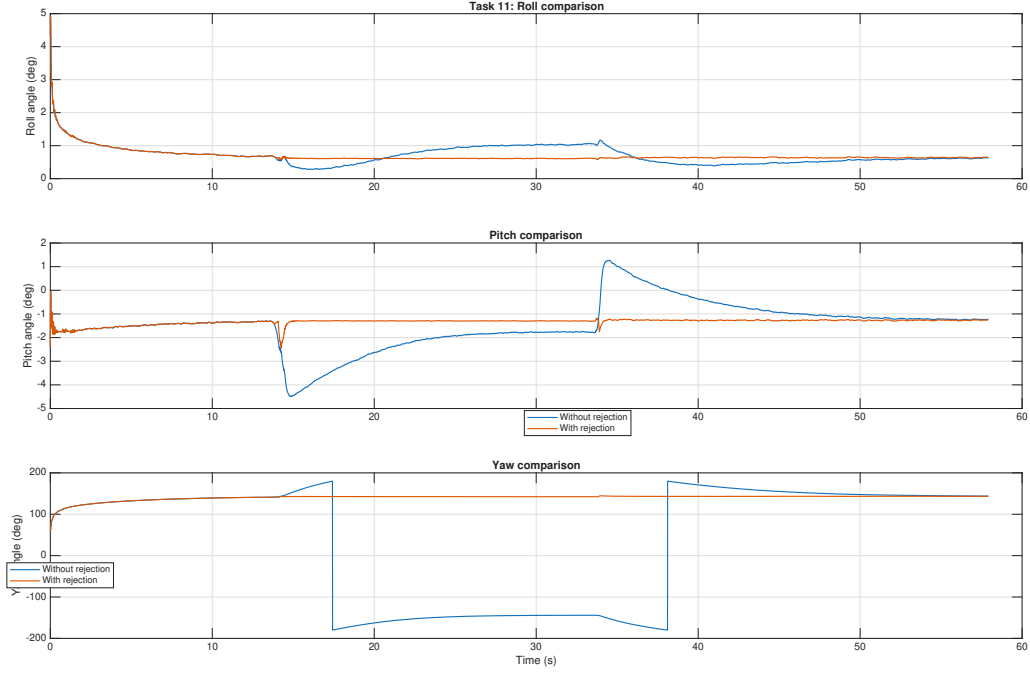


Figure 9: Magnetometer norm and detected magnetic outliers.

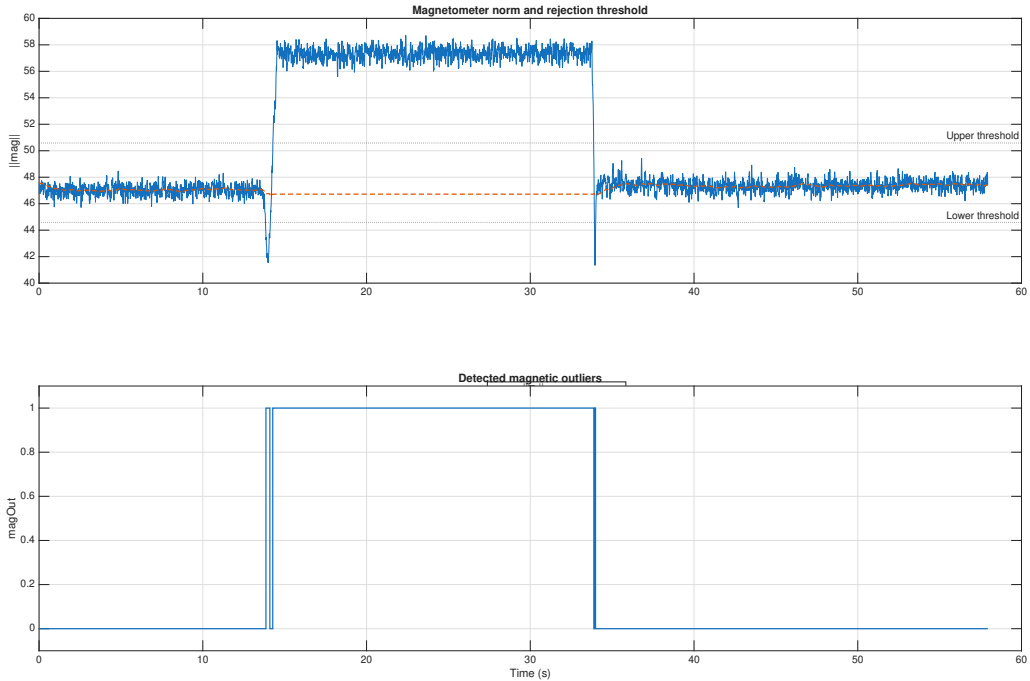


Figure 10: Comparison of attitude estimates with and without magnetic outlier rejection.

To reduce the effect of magnetic disturbances, a magnetometer outlier rejection was added. The detector was based on the magnetometer norm. Let

$$L_k = \|m_k\|.$$

The expected magnetic field strength was estimated slowly using

$$\hat{L}_k = (1 - \alpha)\hat{L}_{k-1} + \alpha L_k.$$

If

$$|L_k - \hat{L}_{k-1}| > \gamma_m,$$

the magnetometer measurement was classified as disturbed and the magnetometer update was skipped.

The outlier detection result is shown in Fig. 9. During the magnetic disturbance interval, the magnetometer norm clearly exceeds the threshold and the detector marks the measurements as outliers. Therefore, the corrupted magnetometer measurements are not used in the EKF update.

Fig. 10 compares the attitude estimates with and without magnetic outlier rejection. Without rejection, the magnetic disturbance causes clear erroneous corrections in roll, pitch, and yaw. With rejection, these disturbances are strongly reduced because the filter relies on the gyroscope prediction and accelerometer update during the disturbed interval, instead of using the corrupted magnetometer measurements.

This shows that the magnetic outlier rejection can effectively reduce the effect of strong magnetic disturbances. However, the method is based only on the magnetic field norm, so it may not detect disturbances that mainly change the magnetic field direction while keeping the norm nearly unchanged.

Task 12

Figure 11 and 12 illustrate the tracking performance under four different sensor combinations.

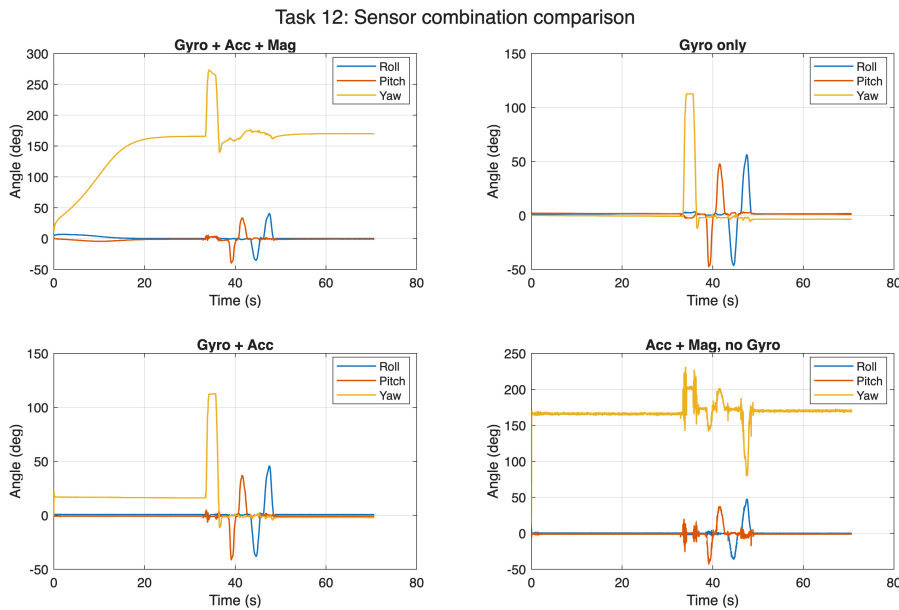


Figure 11: Final filter with four combinations

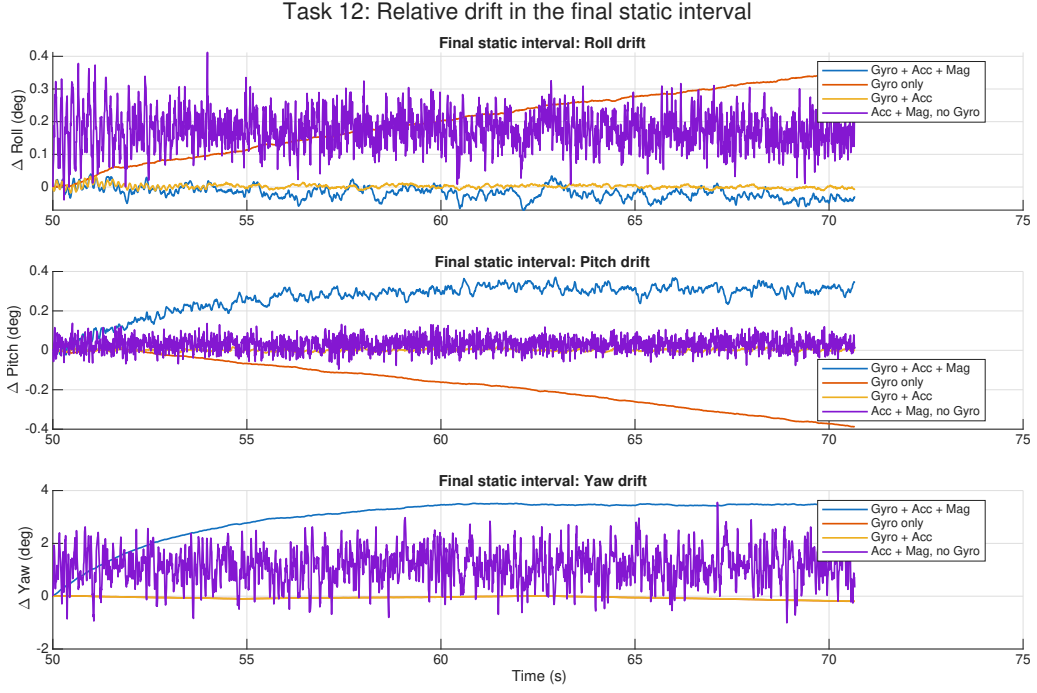


Figure 12: Static relative drift

As verified in Task 8, the yaw angle drifts in the absence of a magnetometer, which is clearly depicted in the results. Furthermore, when relying solely on the gyroscope, all three Euler angles exhibit drift.

By using all of the sensors it was possible to obtain an estimate close to the "Google" orientation. This is illustrated in figure 11 ,12 and 13.

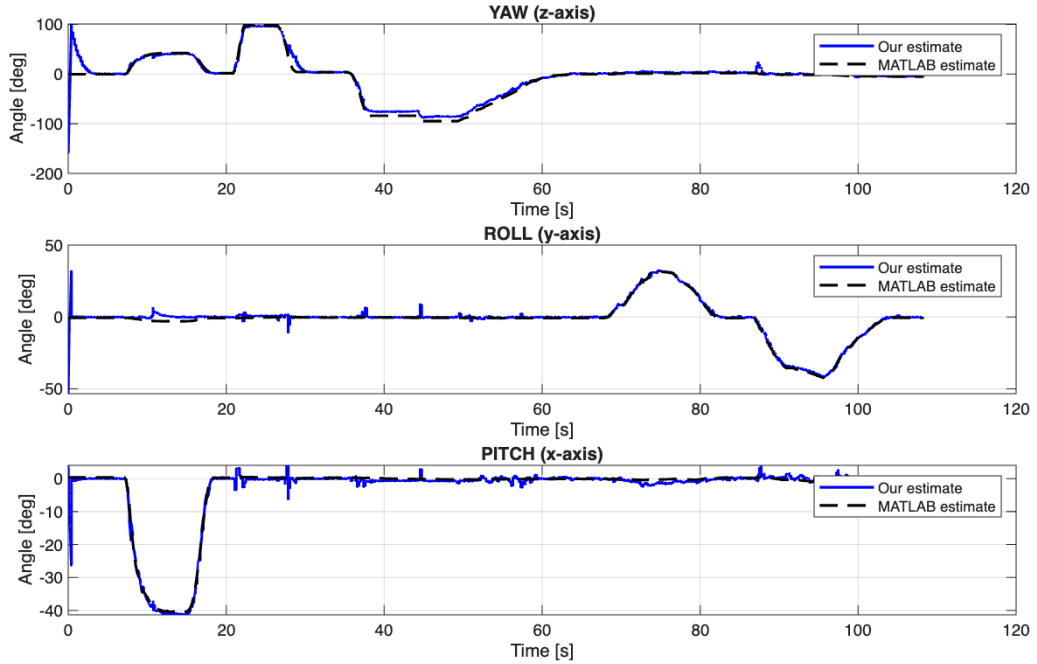


Figure 13: Final filter compared to Mtlab filter

When relying on the accelerometer and magnetometer, the filter manages to track the overall orientation, but the estimation becomes unstable. As observed in figure 11, 13

and 14, the tracking process exhibits jitter and frequent spikes.

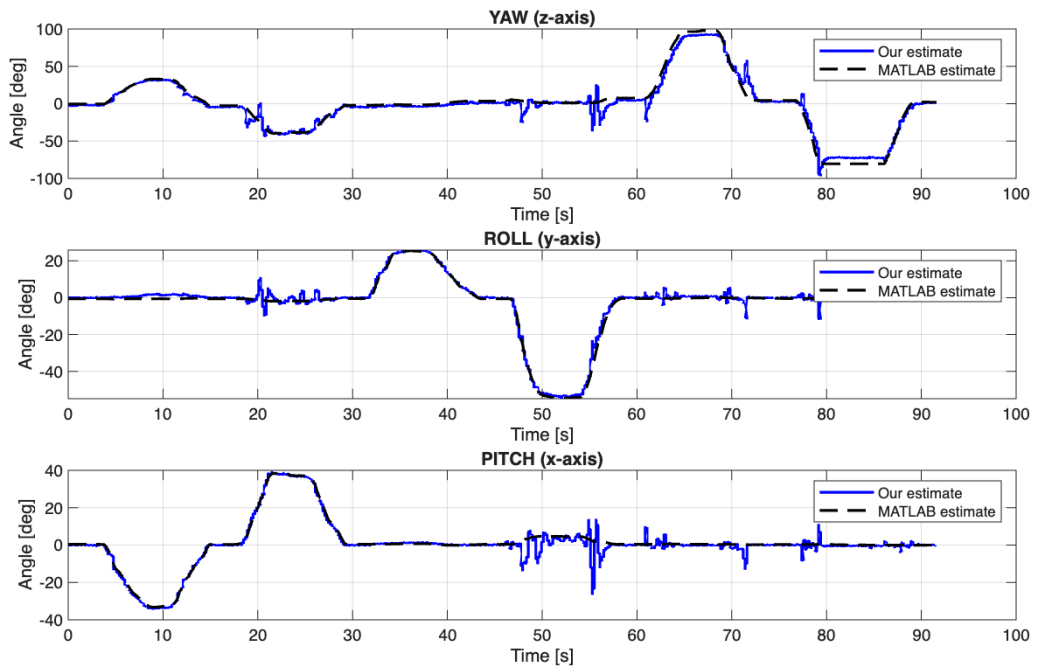


Figure 14: Final filter without gyroscope